
MODULE *Exokernel*

EXTENDS *Naturals, Sequences*

Parameters and constants

CONSTANT *TaskCount, ResourceCount*

VARIABLE *tasks, resources, allocations*

TaskId are modeled as a sequence of task *IDs*, where each task has an index.

$TaskId \triangleq 1 \dots TaskCount$

ResourceId are modeled as a sequence of resource *IDs*, where each resource has an index.

$ResourceId \triangleq 1 \dots ResourceCount$

$TaskState \triangleq \{\text{"waiting"}, \text{"running"}\}$

$Availability \triangleq \{\text{"free"}, \text{"busy"}\}$

Type invariant of the specification

$ExokernelTypeInvariant \triangleq$
 $\wedge tasks \in [TaskId \rightarrow TaskState]$
 $\wedge resources \in [ResourceId \rightarrow Availability]$
 $\wedge allocations \in [TaskId \rightarrow [ResourceId \rightarrow \text{BOOLEAN}]]$

Define *Exokernel* state as a record containing tasks, resources, and allocations

$ExokernelInit \triangleq$
 $\wedge tasks = [t \in TaskId \mapsto \text{"waiting"}]$
 $\wedge resources = [r \in ResourceId \mapsto \text{"free"}]$
 $\wedge allocations = [t \in TaskId \mapsto [r \in ResourceId \mapsto \text{FALSE}]]$

A task can request access to a resource if it's not already in use.

$RequestResource(t, r) \triangleq$
 $\wedge resources[r] = \text{"free"}$
 $\wedge allocations[t][r] = \text{FALSE}$
 $\wedge tasks[t] = \text{"waiting"}$
 $\wedge tasks' = [tasks \text{ EXCEPT } ![t] = \text{"running"}]$
 $\wedge resources' = [resources \text{ EXCEPT } ![r] = \text{"busy"}]$
 $\wedge allocations' = [allocations \text{ EXCEPT } ![t][r] = \text{TRUE}]$

$FreeResource \triangleq \text{CHOOSE } r \in ResourceId : resources[r] = \text{"free"}$

$OccupyResource \triangleq$
 $\wedge \exists r \in ResourceId : resources[r] = \text{"free"}$
 $\wedge \text{LET } res \triangleq \text{CHOOSE } r \in ResourceId : resources[r] = \text{"free"}$
 $\text{IN } resources' = [resources \text{ EXCEPT } ![res] = \text{"busy"}]$
 $\wedge \text{UNCHANGED } \langle tasks, allocations \rangle$

A task can release a resource when it has finished using it.

$$\begin{aligned}
\text{ReleaseResource}(t, r) &\triangleq \\
&\wedge \text{allocations}[t][r] = \text{TRUE} \\
&\wedge \text{tasks}[t] = \text{"running"} \\
&\wedge \text{tasks}' = [\text{tasks} \text{ EXCEPT } ![t] = \text{"waiting"}] \\
&\wedge \text{resources}' = [\text{resources} \text{ EXCEPT } ![r] = \text{"free"}] \\
&\wedge \text{allocations}' = [\text{allocations} \text{ EXCEPT } ![t][r] = \text{FALSE}]
\end{aligned}$$

The next-state relation defines valid state transitions

$$\begin{aligned}
\text{ExokernelNext} &\triangleq \\
&\exists t \in \text{TaskId}, r \in \text{ResourceId} : \\
&\quad (\text{RequestResource}(t, r) \vee \text{ReleaseResource}(t, r))
\end{aligned}$$

Specification of the overall system behavior

$$\text{ExokernelSpec} \triangleq \text{ExokernelInit} \wedge \Box[\text{ExokernelNext}]_{\langle \text{tasks}, \text{resources}, \text{allocations} \rangle}$$